

=====

[MANDATORY NOTICE]

SYNTHETIC DEMONSTRATION DATA

This document is fictional test data created for the VYASA prototype.

It is not an actual operating procedure, engineering standard,
regulatory requirement, or safety instruction.

All numerical values are example synthetic demonstration values only.

SECTION 1: SCOPE, EQUIPMENT & OPERATING PARAMETERS

1.1 Scope & Equipment Covered

This synthetic procedure governs the condition monitoring, routine operation, and abnormal conditions for multistage centrifugal recycle gas compressors, specifically example equipment unit compressor C-201 and compressor C-202.

1.2 Core Operating Parameters (Synthetic Demonstration Values)

- Example synthetic observation suction pressure: 18.0 bar gauge.
- Example synthetic observation discharge pressure: 45.0 bar gauge.
- Nominal operating shaft speed: 9,800 RPM.

=====

SECTION 2: COMPRESSOR C-202 THRESHOLDS & OPERATING OBSERVATIONS

2.1 Vibration Limits for Compressor C-202

Example / Synthetic Demonstration Values for Compressor C-202:

- Normal baseline vibration: < 3.0 mm/s RMS.
- Example synthetic demonstration warning threshold for compressor C-202: vibration > 4.5 mm/s.
- Example synthetic demonstration critical shutdown trip threshold for compressor C-202: vibration > 7.0 mm/s.

2.2 Temperature & Pressure Operating Observations

- Example synthetic demonstration discharge temperature warning limit: > 120.0 °C.
- Example synthetic demonstration maximum discharge temperature trip: > 135.0 °C.
- Lube oil header supply temperature: example synthetic observation range 40.0 °C to 50.0 °C.

=====

SECTION 3: ABNORMAL-CONDITION INDICATORS & OPERATING OBSERVATIONS

3.1 Documented Compressor Operating Observations

- Surge Detection: Characterized by rapid flow reversals, violent pressure fluctuations, and acoustic rumbling.
- Anti-Surge Controller: Modulates recycle valve to maintain operating point at minimum 10% above surge control line.
- Lube Oil Differential Pressure: Example synthetic observation minimum differential pressure of 1.5 bar across filters.
- Dry Gas Seal Leakage: Example synthetic observation threshold of > 15.0 Nm³/h primary vent flow requires seal examination.